Analyser — User Guide

> **Version:** 1.3 · **Last updated:** May 2026

Analyser is a browser-based tool for inspecting and comparing `.fit` activity files from Garmin devices and other ANT+ sensors. It runs entirely in your browser — no account, no upload, no server. Files are parsed locally.

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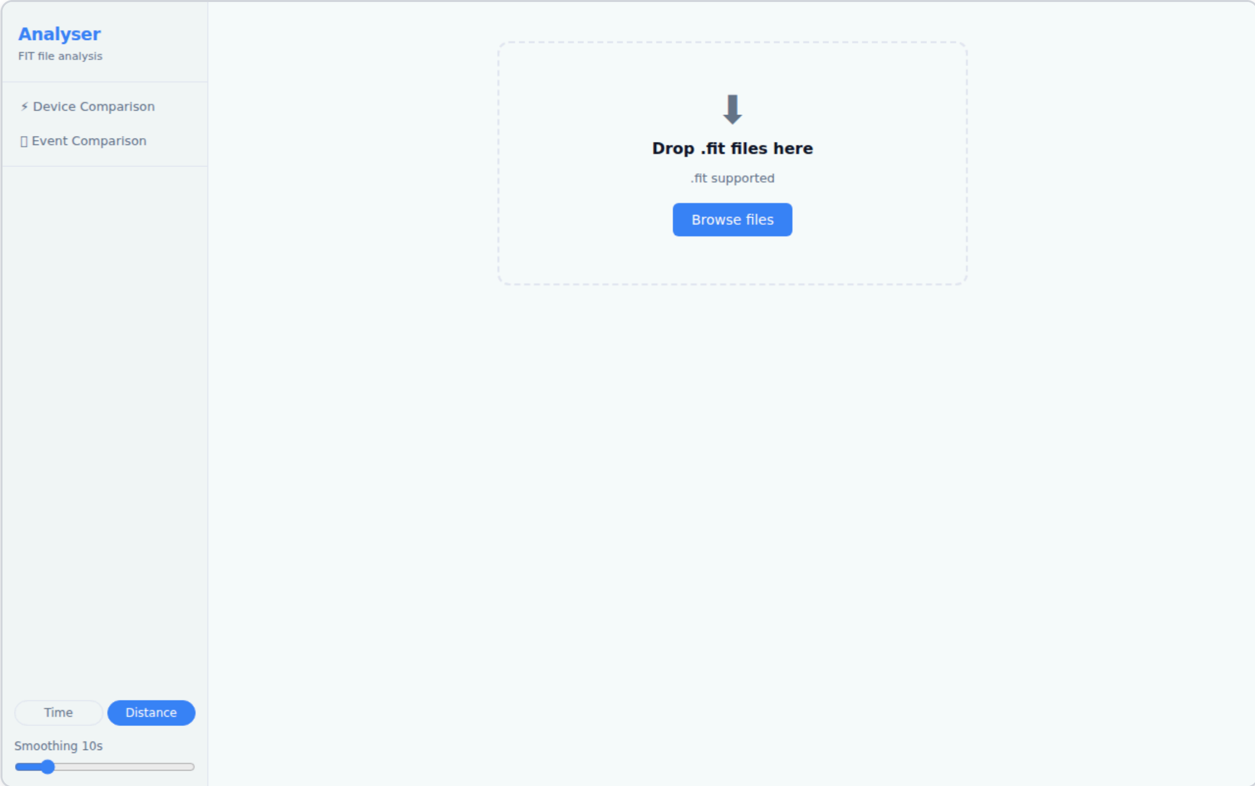
1. Getting Started

Open Analyser in your browser. You are greeted with the landing page, which contains a drop zone and a file picker.



To load a file:

- **Drag and drop** one or more `.fit` files onto the drop zone, or
- Click **Choose files** to open the system file picker.

You can load **up to 6 files** at once. Once the files are parsed, you are taken directly to the **Device Comparison** view.

> **Supported format:** `.fit` (Flexible and Interoperable Data Transfer — the standard format exported by Garmin watches, Wahoo computers, and most ANT+ devices). Files from `.tcx` and `.gpx` sources are not currently supported.

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2. The Interface

After loading a file, the application has two main views:

View	What it shows
⚡ **Device Comparison**	Sensor streams from one or more files displayed side-by-side on a shared time or distance axis
▯ **Event Comparison**	Two or more runs of the same course aligned by distance, showing where time was gained or lost

The **Device Comparison** view (described in sections 3–4) is the primary view and opens automatically when you load a file.

Navigation

On a **desktop** browser, the two views are accessible from the left sidebar. Click ⚡ **Device Comparison** or ▯ **Event Comparison** to switch modes.

On a **tablet or phone**, the sidebar is hidden by default. Tap the ☰ **menu button** in the top-left corner to slide out the navigation drawer. Tap any item or tap outside the drawer to close it. You can also swipe left across the drawer to close it.

On a **phone** (narrow screen), a **bottom navigation bar** appears at the foot of the screen with the same ⚡ Device and ▯ Event buttons for quick switching without opening the drawer.

Toolbar Controls

On tablet and phone, the toolbar controls (Device Toggle Bar, Channel Toggle Bar, Fine-tune timing) are hidden behind a collapsible panel. Tap the **Devices & Options** (or **Channels & Options**) button to expand or collapse the controls. On desktop the controls are always visible.

Theme

The **Theme** control is in the bottom of the sidebar (or navigation drawer), between the X-axis toggle and the Smoothing slider.

Option	Behaviour
▯ **Sys** (default)	Follows your operating system's dark/light preference
☀ **Lit**	Forces the light theme regardless of OS setting
▯ **Drk**	Forces the dark theme regardless of OS setting

Your choice is saved in your browser and restored the next time you open Analyser.

3. Device Comparison — Single File

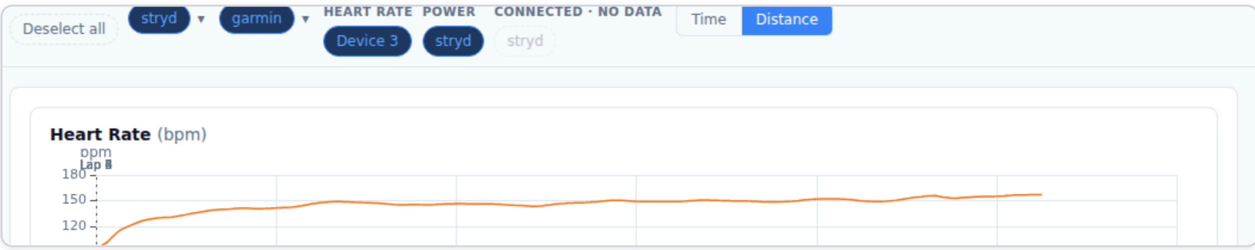
Loading a single file opens the Device Comparison view with all detected sensor streams active.

The view is divided into:

- **Left sidebar** — navigation and file controls
- **Device Toggle Bar** — sensors detected in the file, grouped by channel
- **Tab panel** — Charts, Summary, Mean/Max, and Map tabs

3.1 Device Toggle Bar

The Device Toggle Bar sits at the top of the content area and lists every sensor stream detected in the loaded file.



Key concepts:

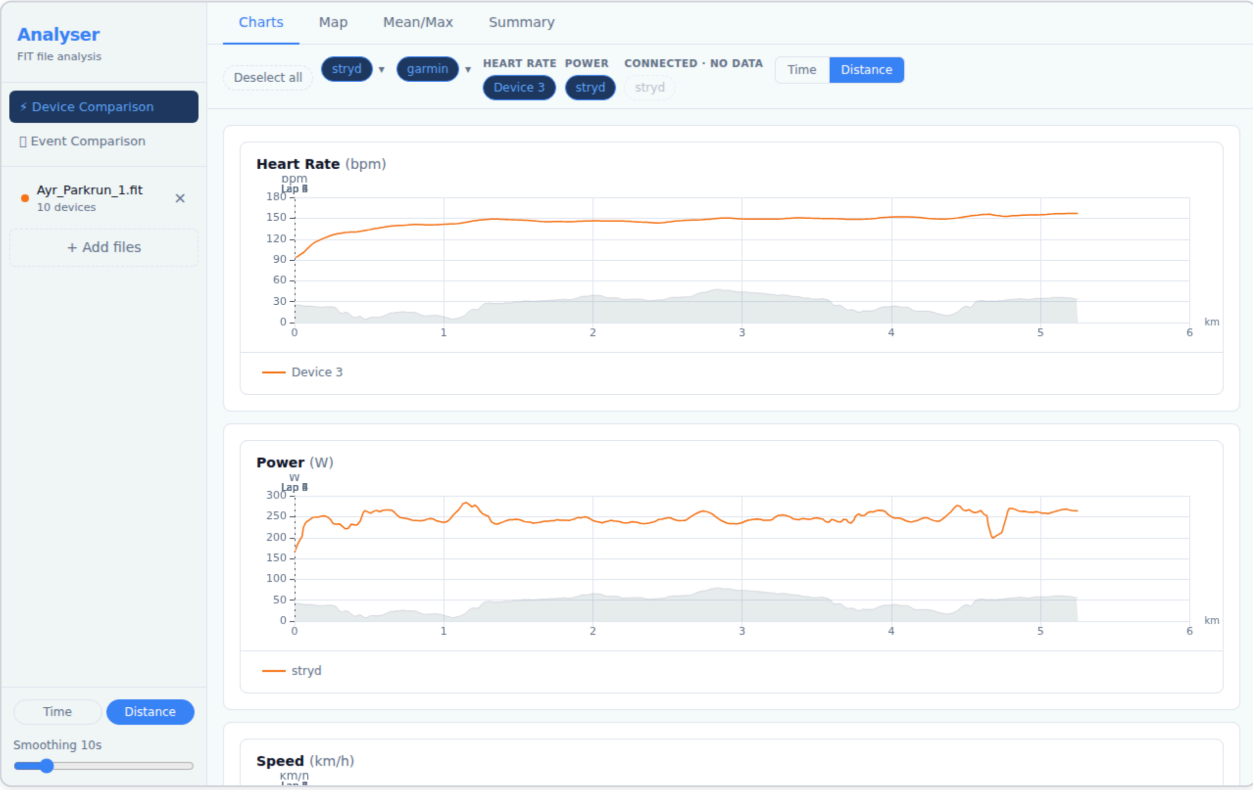
- Each **pill** represents one device contributing to a channel (e.g. a Polar H10 contributing Heart Rate).
- **Active** devices (highlighted in blue) are included in all charts and statistics. Click a pill to toggle it on or off.
- Channels with a ✦ indicator have data from two or more devices — these are directly comparable.
- **Multi-metric devices** (devices recording 3+ channels, such as a Garmin watch) appear as a single expandable pill. Click the ▾ arrow to see the individual channels.
- Devices that were connected but recorded no data are shown as dashed, semi-transparent pills and cannot be toggled.

Renaming a device:

Double-click any device pill to rename it. Type the new label and press **Enter** to confirm, or **Escape** to cancel. Labels are saved locally and will be recognised the next time a file from the same ANT+ device is loaded.

3.2 Charts Tab

The Charts tab displays time-series data for every active channel.



Reading the charts:

- Each chart plots one channel (e.g. Heart Rate, Power, Pace).
- The **X axis** is either elapsed time or distance — toggle between them with the **Time / Distance** buttons at the top.
- Multiple devices contributing to the same channel each appear as a separate line.
- **Pace** uses an inverted Y axis — faster pace (lower min/km) appears higher on the chart, which matches the intuition of "going faster".

Interacting with the charts:

Action	Effect
Hover	Tooltip shows exact values across all series at that position
Click + drag	Zoom in on a time/distance range
Double-click	Reset zoom
Scroll wheel	Zoom in/out (all charts stay synchronised)

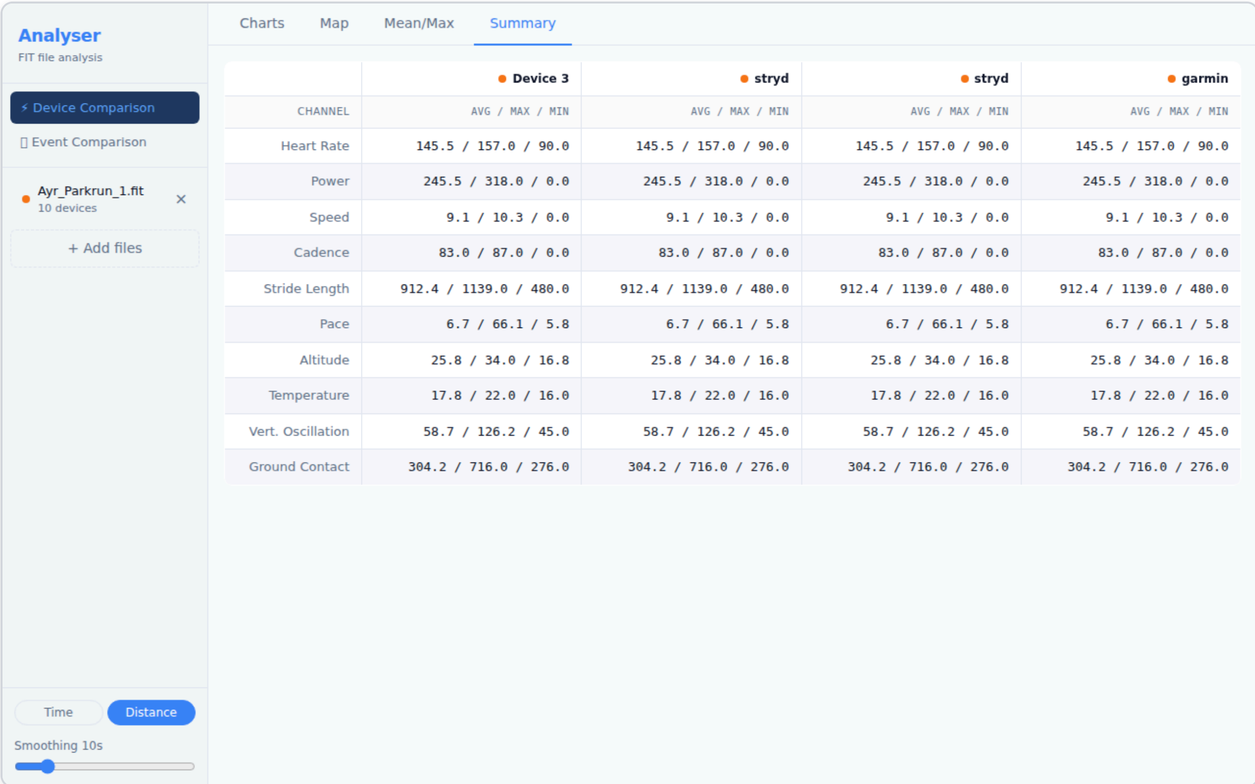
Lap markers appear as faint vertical lines across all charts. Hover a marker to see the lap number and split time.

Smoothing:

A smoothing control lets you apply a rolling average to reduce sensor noise. Increase the window (in seconds) to smooth out spikes; set it to 1 s for raw data. Smoothing applies to all channels simultaneously.

3.3 Summary Tab

The Summary tab shows aggregate statistics for every active device and channel.



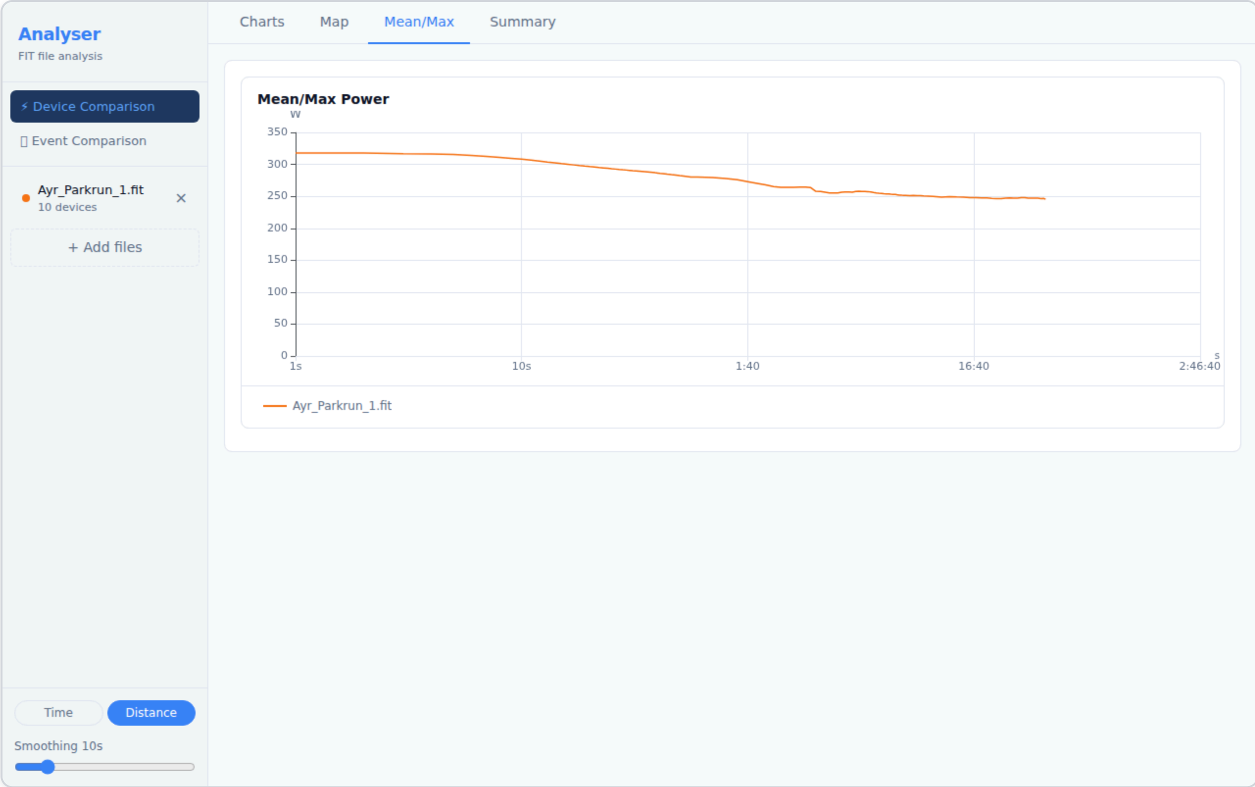
For each channel, the table shows:

Column	Meaning
Device	The sensor that recorded this channel
Avg	Time-weighted average over the active portion of the activity
Max	Peak value recorded
Total	Cumulative value (distance, calories) — shown where applicable

Toggling devices in the Device Toggle Bar updates the Summary table immediately.

3.4 Mean/Max Tab

The Mean/Max tab plots the **best average power** (or other metric) over every possible duration, from one second up to the full activity length.

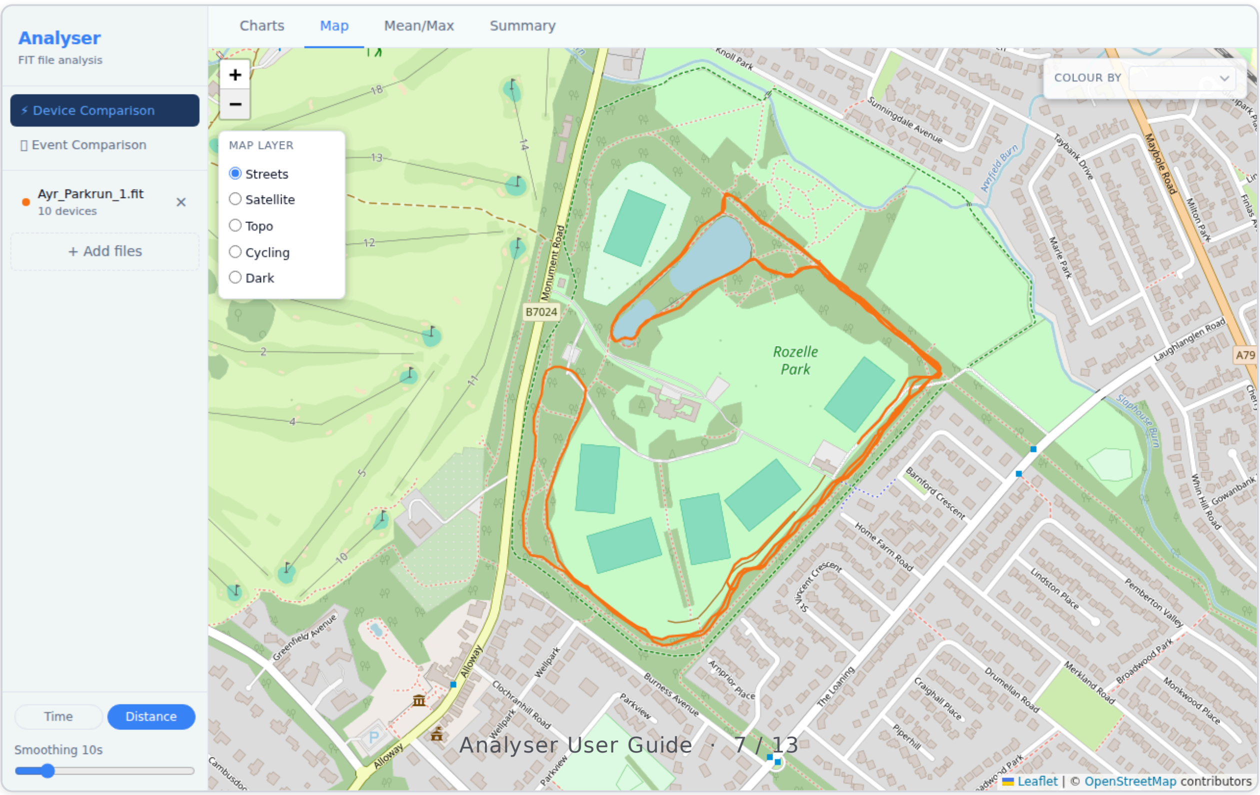


This is the standard "Critical Power curve" or "Power Duration curve" used in cycling and running:

- The Y axis is the best average power achieved for the duration on the X axis.
- A high, flat curve indicates sustained high output; a steep drop-off indicates strength over short durations but limited endurance.
- When multiple files are loaded, each file produces its own curve — load two sessions to compare fitness progress.

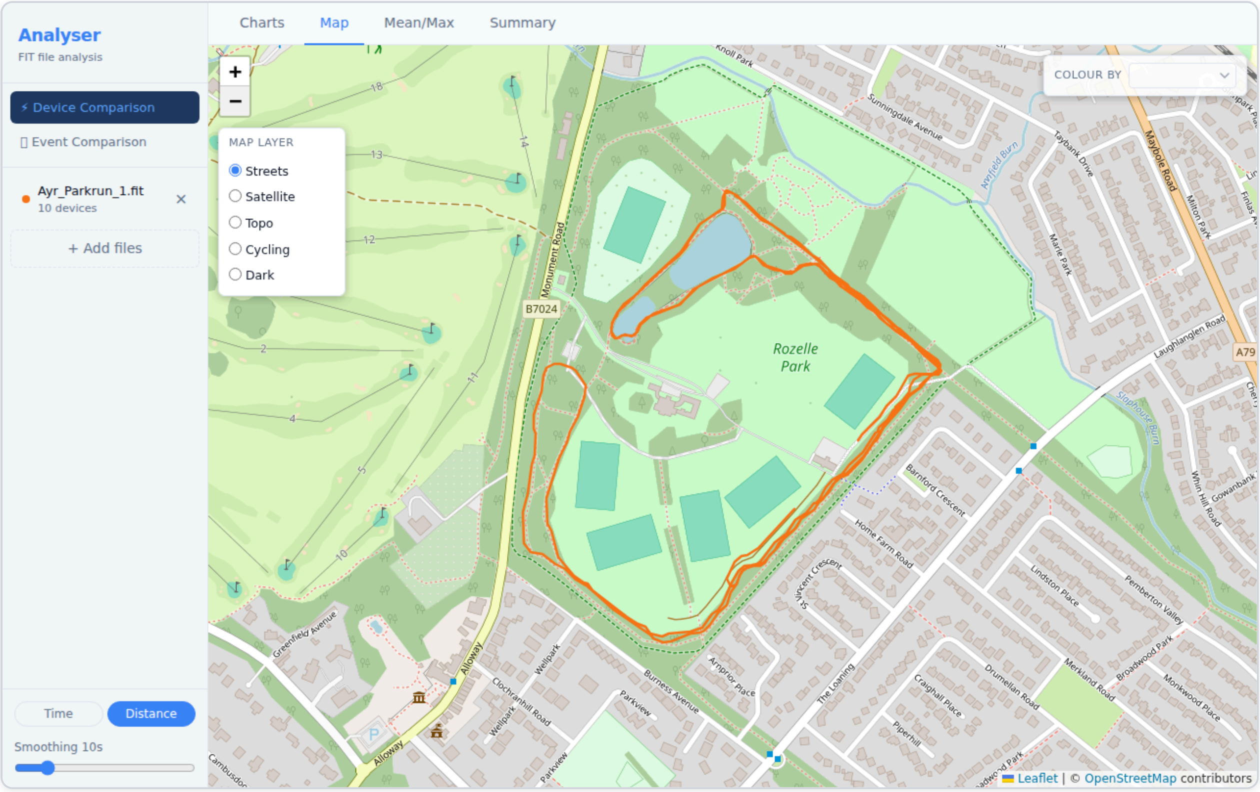
3.5 Map Tab

The Map tab renders the GPS route recorded in the activity.



Tile layers:

A **Map Layer** panel is always visible in the **top-left** corner of the map. Click any layer name to switch the base map:



Layer	Best for
Streets (default)	Road runs, orientating yourself
Satellite	Seeing the actual terrain
Topo	Hilly routes — contour lines visible
Cycling	Bike-specific road markings
Dark	Low-glare, high-contrast viewing

Metric colouring:

Use the **Colour by** dropdown in the **top-right** of the map to shade the route by a metric such as pace, heart rate, or power. The colour gradient runs from blue (low values) to red (high values). A legend appears in the bottom-right showing the value range.

Hover interaction:

Hovering over the map route shows a crosshair on all open charts, letting you pinpoint exactly what your metrics were at a specific GPS location.

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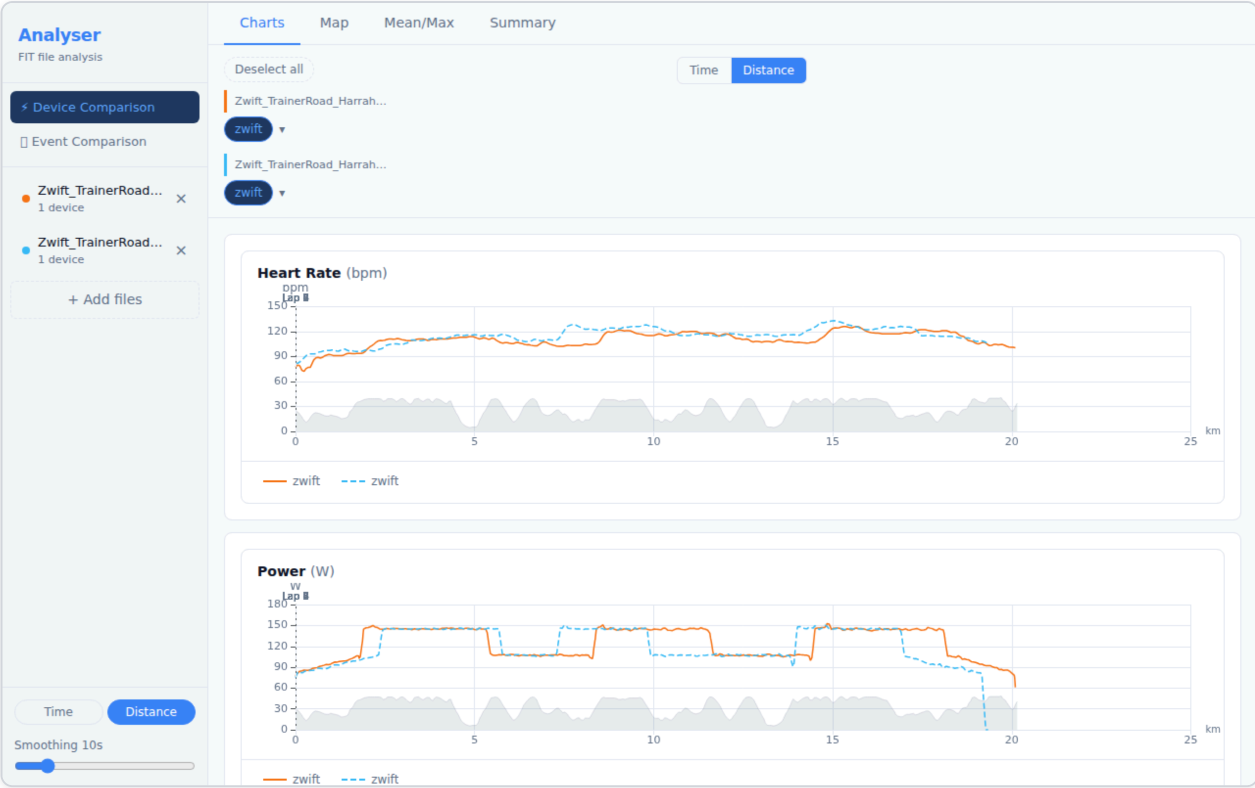
4. Device Comparison — Multiple Files

Analyser supports loading up to **6 files simultaneously**, enabling you to compare sensor data across different activities or devices.

4.1 Loading Multiple Files

Select multiple `.fit` files in the file picker (hold Shift or Cmd/Ctrl to select more than one), or drag and drop multiple files onto the drop zone at once.

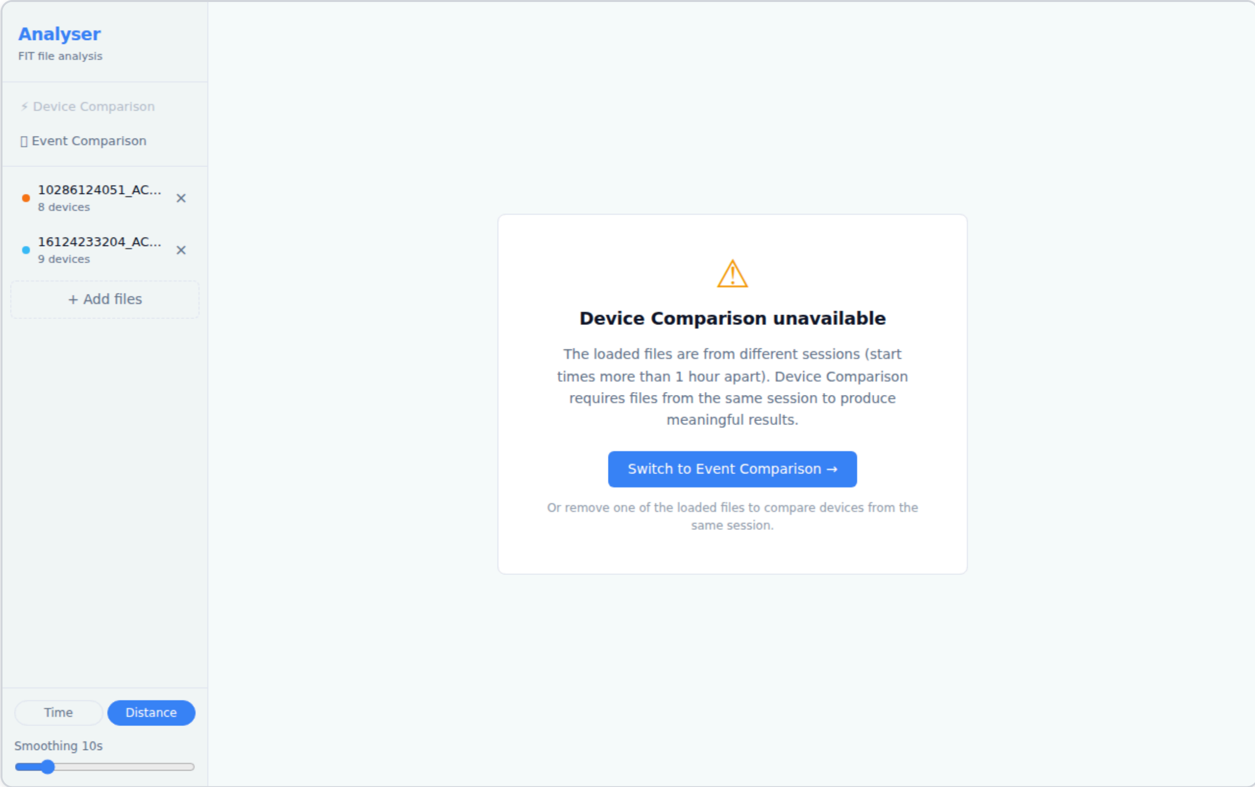


> **Note:** If the loaded files are from different sessions (start times more than 1 hour apart), Device Comparison will be unavailable — see section 4.2.

4.2 Different-Session Files

Device Comparison is designed for files recorded **during the same session** — for example, two devices worn simultaneously on the same run. Comparing files from completely different activities (different days, different events) produces charts with no meaningful relationship between the data streams.

If you load two or more files whose start times are more than 1 hour apart, Analyser detects this and shows an explanation panel instead of charts:



The ⚡ **Device Comparison** button in the sidebar also appears greyed-out with a tooltip explaining why it is unavailable.

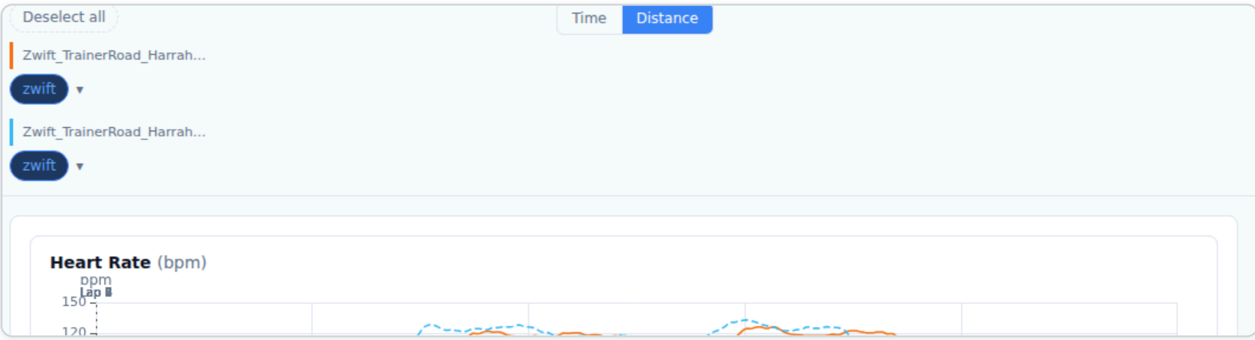
What to do:

- Click **Switch to Event Comparison →** to use the ▯ Event Comparison view, which is designed for comparing runs of the same course across different sessions.
- Or remove one of the loaded files (using the × button in the sidebar) until the remaining files are from the same session.

Once the remaining files overlap in time (within 1 hour), Device Comparison becomes available again automatically.

4.3 Multi-File Device Bar

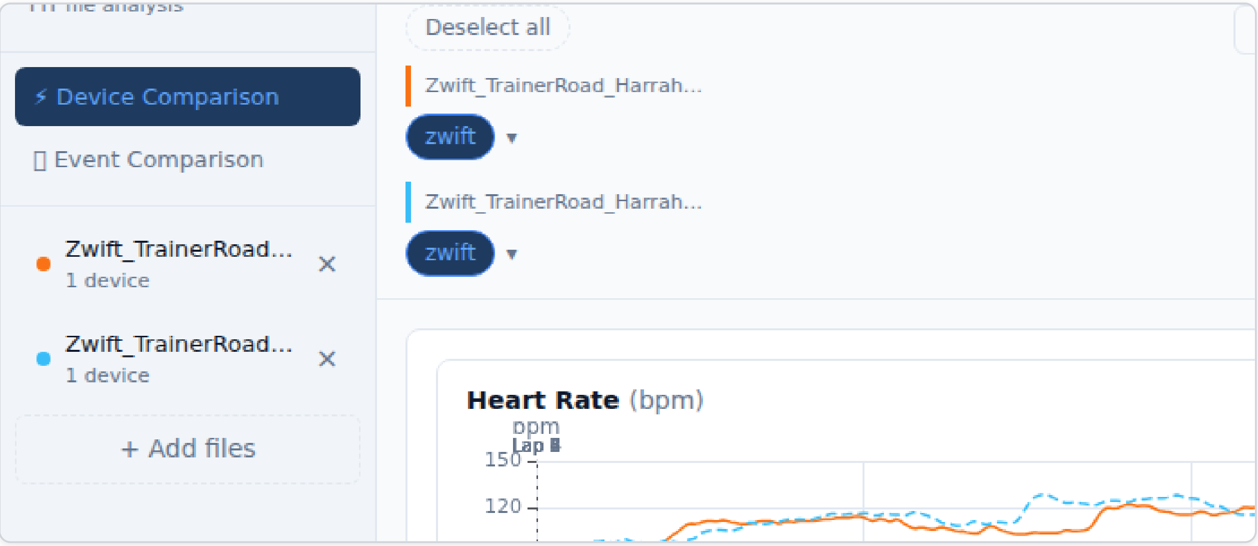
In multi-file mode the Device Toggle Bar groups sensors by their source file, with a coloured left-border header for each file.



- Each file gets a unique **colour** (blue, orange, green, purple, …). All chart series from that file share the same colour.
- Channels with data from two or more files display the ✦ comparable indicator — these are the most valuable channels to compare.
- Use the **Select all / Deselect all** button to quickly show or hide everything.

4.4 Fine-Tuning Timing

Even when files are from the same session, GPS clocks sometimes drift by a few seconds. The **Fine-tune timing** control lets you shift any file forwards or backwards in time to align the traces precisely.



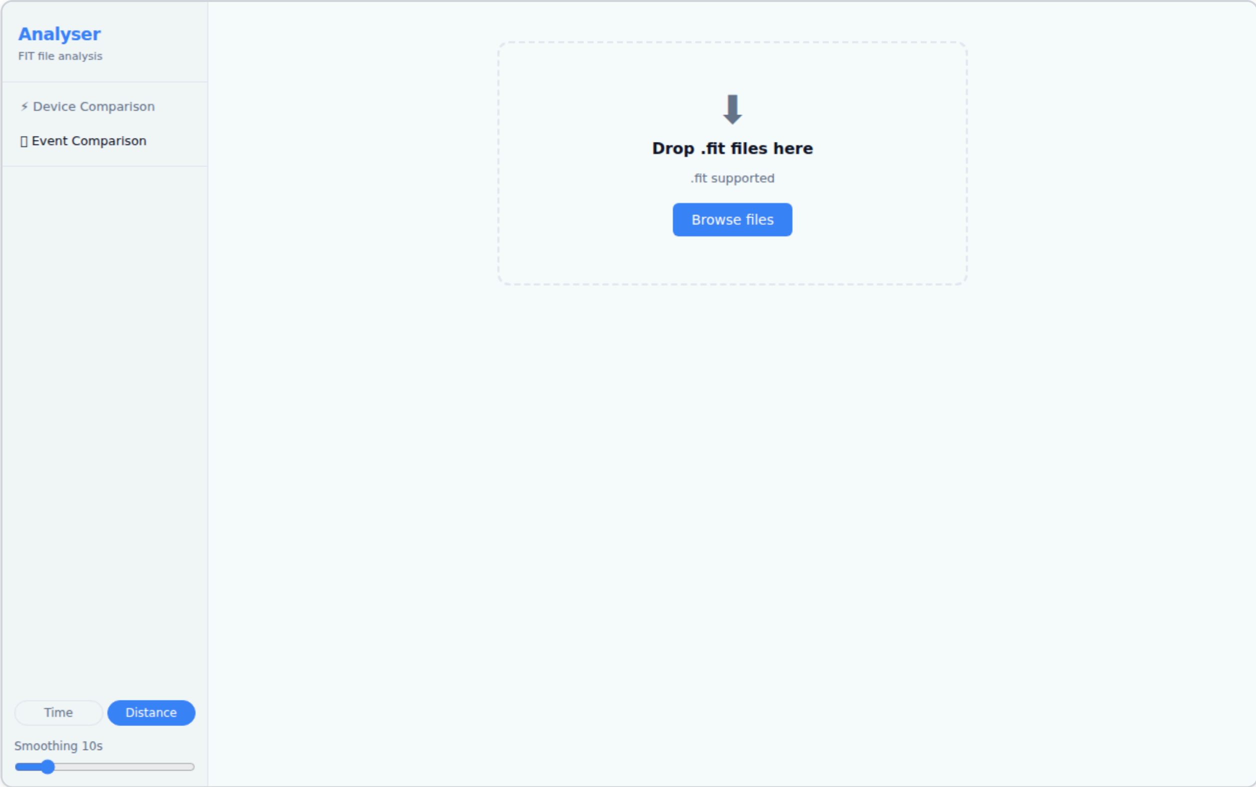
Click the ▯ **Fine-tune timing** button (below the Device Toggle Bar, visible in multi-file + time-axis mode) to expand the panel. This control is only relevant when files are from the **same session** — if they are from different sessions, the gate panel is shown instead (see section 4.2).

Control	Function
−10 / −1 buttons	Shift the file 10 s or 1 s earlier
+1 / +10 buttons	Shift the file 1 s or 10 s later
Numeric input	Type an exact offset in seconds
↺ (Reset)	Return to the auto-computed offset

The first loaded file is always the **reference** (offset = 0). All other files are shifted relative to it. The offset is shown in the **N adjusted** badge on the toggle button.

5. Event Comparison

The **Event Comparison** view (▯) is designed for comparing multiple runs of the same course — for example, several parkruns on the same route — aligned by **distance** to show exactly where time was gained or lost.



Load two or more `.fit` files, then switch to the Event Comparison tab. The view provides:

- **Time delta plot** — cumulative seconds gained or lost versus distance. Positive values mean you were ahead of the reference run; negative values mean you were behind.
- **Overlaid time-series charts** — pace, speed, heart rate, and power for all files on a shared distance axis.
- **Segment bar chart** — per-kilometre or per-lap time difference, highlighting your best and worst segments.

> **Tip:** Select a reference run using the controls in the sidebar. The time delta is calculated relative to the reference. Try different reference runs to understand where you consistently gain or lose time.

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6. Reference

Keyboard Shortcuts

Key	Action
Double-click a chart	Reset zoom
Enter (in rename input)	Confirm device rename
Escape (in rename input)	Cancel device rename
Enter (in offset input)	Confirm time offset

Supported Sensor Types

Sensor	Channels recorded
Heart Rate Monitor (HRM)	Heart Rate
Power Meter (bike)	Power, Left/Right balance
Stryd Running Power	Power, Form Power, Ground Contact Time, Stride Length
Core Body Temperature	Core Temperature, Skin Temperature
Speed/Cadence Sensor	Speed, Cadence
Running Dynamics Pod	Vertical Oscillation, Ground Contact Time, Stride Length
Watch / Primary device	All remaining channels: GPS, Altitude, Speed, Pace, Temperature

File Limits

- **Maximum files per session:** 6
- **Supported format:** `.fit` (ANT/Garmin FIT protocol)
- All parsing happens **locally in your browser** — no data is uploaded to any server.

Troubleshooting

Symptom	Likely cause	Fix
File loads but no charts appear	No sensor data in the file	Check the Device Toggle Bar — all devices may be deselected or no-data
Device Comparison shows "unavailable" panel	Files are from different sessions (start times >1 hour apart)	Use **Event Comparison** (▯) to compare runs of the same course, or remove the out-of-session file
Two same-session files don't align on the time axis	GPS clock drift between devices	Adjust offsets using the Fine-tune timing panel (section 4.4)
Map shows no route	FIT file has no GPS data	Some indoor activities (turbo trainer, treadmill) do not record GPS
Pace chart looks upside-down	Expected — faster pace (lower min/km) is plotted higher	This is correct; it matches the intuition of "going faster = higher on chart"
Device name shows as "Device N"	ANT+ device ID not yet labelled	Double-click the pill to rename it; the label is saved for future sessions

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Screenshots taken using real activity files: Ayr Parkrun and Zwift/TrainerRoad sessions.